2025 2nd International Conference on Research Methodologies in Knowledge Management, Artificial Intelligence and Telecommunication Engineering (RMKMATE) | 979-8-3315-9848-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/RMKMATE64874.2025.11042816

MEDFIT-LLM: Medical Enhancements through Domain-Focused Fine Tuning of Small Language Models

Aditya Karnam Gururaj Rao[1#], Arjun Jaggi[2*],Sonam Naidu[3^]

[#]Zefr Inc, LA, USA

[*]HCLTech, LA, USA

[^]LexisNexis, USA

[1]akarnam37@gmail.com

[2]arjunjaggi@gmail.com

[3]sonam.naidu20@gmail.com

***Abstract*— This research explores the efficacy of fine-tuning small Large Language Models (LLMs) for AI-based healthcare chatbots. We present a novel approach to dataset creation using synthetic data generation and fine-tuning of various LLMs. The study compares the performance of base models against their fine-tuned counterparts using a carefully curated dataset of healthcare-related questions and answers. Our methodology involves the use of phi 4 for synthetic data generation, supplemented with domain-specific questions derived from existing research. We employ LORA fine-tuning on MLX for four distinct models: Gemma 2 9B 4bit, LLama 3.2 3B Instruct, Mistral 7B, and Qwen2 7B Instruct 8 bit. The results demonstrate the potential of fine-tuning in enhancing the performance of LLMs in specialized healthcare contexts, contributing to the ongoing development of more accurate and reliable AI-powered healthcare chatbots. The code is available at https://github.com/adityak74/medfit-llm.**

Keywords— healthcare chatbots, fine-tuning, small language models, synthetic data generation, lora, mlx, gemma, llama, mistral, qwen, response efficiency, direct answer improvement, healthcare ai, medical information dissemination, domain-specific training

I. Introduction

The integration of Large Language Models (LLMs) in healthcare applications has shown tremendous potential in revolutionizing patient care, medical research, and healthcare information dissemination [9], [12], [13]. As the demand for accurate and context-aware AI-powered healthcare solutions grows, there is an increasing need for specialized chatbots that can provide reliable medical information and support [2], [10]. This research focuses on enhancing the capabilities of AI healthcare chatbots through the fine-tuning of small LLMs, addressing the challenges of domain-specific knowledge and contextual understanding in medical conversations.

The primary objective of this study is to evaluate the effectiveness of fine-tuning various small LLMs for healthcare-specific tasks. By comparing the performance of base models against their fine-tuned versions, we aim to demonstrate the potential improvements in accuracy and relevance of responses in healthcare contexts. This research contributes to the ongoing efforts to develop more reliable and efficient AI-powered healthcare communication tools.

II. Related Work

Recent advancements in LLMs have sparked considerable interest in their application to healthcare [1], [9], [10]. This research proposed a comprehensive methodology for enhancing the reliability of healthcare conversational agents, focusing on the specific needs of Hajj pilgrims [2]. Their approach leveraged domain-specific fine-tuning techniques and synthetic data augmentation strategies to optimize performance in delivering contextually relevant healthcare information.

Similarly, this other research paper conducted a comparative analysis of various LLMs for deployment in women's healthcare mobile applications [4]. Their study evaluated models such as BERT, DistilBERT, Roberta, DialoGPT, TinyLLAMA, and GPT-2, highlighting the potential of text-generation models in delivering engaging responses and enhancing health awareness among women.

The work of in PharmaLLM demonstrated the potential of LLMs in specialized medical tasks such as medicine prescription. Their research underscores the importance of domain-specific training and fine-tuning in developing reliable healthcare AI agents [3].

III. Methodology

A. Dataset Creation

Authorized licensed use limited to: The University of Toronto. Downloaded on August 05,2025 at 10:52:58 UTC from IEEE Xplore. Restrictions apply.

Our approach to dataset creation involved a novel combination of synthetic data generation and domain-specific question formulation. We utilized Phi-4 to generate a synthetic dataset of 10000 healthcare-related questions and answers, focusing on the future impact of LLMs in healthcare. After filtering for duplicates, the final dataset comprised 6444 unique questions.

The dataset was split into training, testing, and validation sets, with 5155 samples for training, 645 for testing, and 644 for validation. This distribution ensures a robust evaluation of the models' performance on unseen data while maintaining a substantial training set.

B. *Model Selection and Fine Tuning*

We selected four LLMs for this study, representing a range of architectures and sizes:

1. Gemma 2 9B 4bit [14]
2. LLama 3.2 3B Instruct [11]
3. Mistral 7B Instruct v0.3 [15]
4. Qwen2 7B Instruct 8 bit [16]

The fine-tuning process employed LORA (Low-Rank Adaptation) on MLX, a technique that allows for efficient adaptation of pre-trained models to specific tasks [7], [8], [18].

TABLE I
QUANTITATIVE COMPARISON OF BASE AND FINE-TUNED MODELS

Model	Size	Direct Answer (%)	Generation Time (s)	Length	Formatting Changes (%)
Llama-3.2-3B	3B	6.0 → 36.0 (+30.0)	6.95 → 7.06 (+1.6%)	+2.84%	Numbered Lists: +18.00, Bold Text: -34.00
Mistral-7B	7B	10.0 → 30.0 (+20.0)	15.46 → 12.02 (-22.2%)	-22.64%	Numbered Lists: -4.00, Bold Text: +12.00
Gemma-2-9B	9B*	0.0 → 0.0 (0.0)	6.24 → 5.95 (-4.7%)	-2.72%	Numbered Lists: +6.00, Bold Text: 0.00
Qwen2-7B	7B*	42.0 → 42.0 (0.0)	7.91 → 7.25 (-8.3%)	-9.06%	Numbered Lists: -6.00, Bold Text: -10.00

IV. EVALUATION

The evaluation phase involved generating answers to the 50 questions in the validation set using both the base models and their fine-tuned counterparts. This comparative analysis aims to quantify the improvements achieved through fine-tuning and assess the models' performance in healthcare-specific contexts.

V. RESULTS AND DISCUSSION

Table I presents a comprehensive comparison of all key metrics across the four models, providing a quantitative overview of the changes observed after fine-tuning. The following subsections analyze these results in greater detail. Our comparative analysis of base models and their fine-tuned counterparts reveals significant differences in performance across multiple dimensions. These results provide valuable insights into how fine-tuning impacts different LLM architectures in healthcare contexts [5], [7], [8].

A. *Direct Answer Improvement*

Fine-tuning significantly improved the direct answering capabilities of specific models, as shown in Fig. 1. The most dramatic improvement was observed in Llama-3.2-3B, which showed a 30 percentage point increase in direct answer rate (from 6.0% to 36.0%). Similarly, Mistral-7B exhibited a substantial improvement of 20 percentage points (from 10.0% to 30.0%). In contrast, Gemma-2-9B showed no improvement in direct answering capability, while Qwen2-7B maintained its already high direct answer rate (42.0%) after fine-tuning.

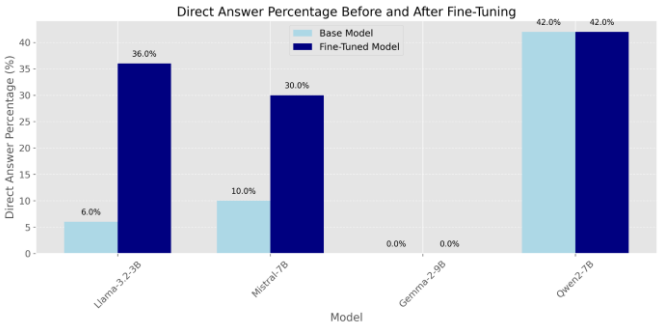

Figure 1. Direct Answer Percentage Before and After Fine-Tuning

Authorized licensed use limited to: The University of Toronto. Downloaded on August 05,2025 at 10:52:58 UTC from IEEE Xplore. Restrictions apply.

B. Generation Efficiency

The impact of fine-tuning on generation time varied considerably across models. Mistral-7B demonstrated the most substantial efficiency gain with a 22.24% reduction in generation time (from 15.46s to 12.02s). Both Gemma-2-9B and Qwen2-7B also showed modest improvements, with 4.66% and 8.25% reductions in generation time, respectively. Interestingly, Llama-3.2-3B was the only model that showed a slight increase in generation time (1.60%), suggesting a potential trade-off between response quality and efficiency for this particular model.

C. Response Structure and Formatting

Fine-tuning had a pronounced effect on the structural elements of model responses [5]. Llama-3.2-3B exhibited the most substantial changes in formatting, with an 18.00% increase in the use of numbered lists accompanied by a 34.00% decrease in bold text usage and a 12.00% reduction in bullet points. These changes suggest a shift toward more organized and systematically structured responses. Mistral-7B showed a contrasting pattern with a 12.00% increase in bold text usage while slightly decreasing numbered lists usage. Gemma-2-9B and Qwen2-7B demonstrated more modest structural changes, with Gemma-2-9B increasing numbered lists by 6.00% and Qwen2-7B reducing both numbered lists and bold text usage by 6.00% and 10.00% respectively.

D. Response Length and Content

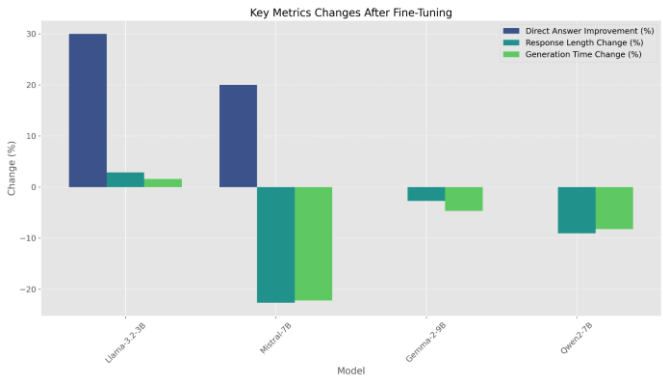

Figure 2. Key Metrics Changes After Fine-Tuning

The change in response length after fine-tuning varied significantly across models (Fig. 2). Llama-3.2-3B was the only model that produced longer responses after fine-tuning, with a 2.84% increase in response length. In contrast, Mistral-7B showed the most substantial decrease in response length at 22.64%, suggesting a shift toward more concise responses. Gemma-2-9B and Qwen2-7B also produced more concise responses after fine-tuning, with decreases of 2.72% and 9.06% respectively.

E. Overall Model Improvements

Our comprehensive scoring system, which weighted various performance metrics including direct answer improvement, response structure, generation efficiency, and question repetition reduction, revealed significant differences in overall improvement across models (Fig. 3). Llama-3.2-3B demonstrated the highest overall improvement score (108.2), followed by Mistral-7B (83.6). Gemma-2-9B and Qwen2-7B showed considerably lower improvement scores of 12.0 and 1.8, respectively.

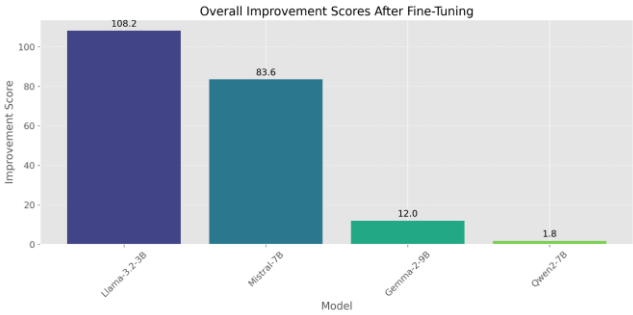

Figure 3. Overall Improvement Scores After Fine-Tuning

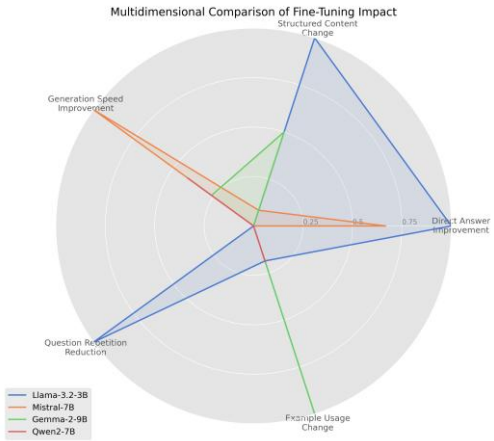

Figure 4. Multidimensional Comparison of Fine-Tuning Impact (Radar Chart)

The radar chart visualization (Fig. 4) provides a multidimensional view of these improvements, highlighting how each model excelled in different aspects. Llama-3.2-3B showed balanced improvements across direct answers, structured content, and question repetition reduction. Mistral-7B excelled primarily in generation speed improvement, while also making gains in direct answering. Gemma-2-9B and Qwen2-7B showed more modest improvements across all dimensions, with Gemma-2-9B performing slightly better in structured content organization.

F. Exemplary Response Improvements

Qualitative analysis of response pairs revealed notable patterns in improvement. In Llama-3.2-3B, fine-tuning resulted in

Authorized licensed use limited to: The University of Toronto. Downloaded on August 05,2025 at 10:52:58 UTC from IEEE Xplore. Restrictions apply.

responses that began with direct answers followed by concise explanations, replacing the previous pattern of beginning with general information. For example, when asked "Can AI tools provide real-time feedback during clinical procedures to improve outcomes?", the base model started with general information while the fine-tuned model began with "Yes, AI tools can offer real-time feedback..." followed by specific details.

Mistral-7B demonstrated the most dramatic transformations in response style, with the fine-tuned model reducing verbose introductions by 60-70% while maintaining essential information. For healthcare questions like "Can patients receive instant feedback from AI health apps?", the fine-tuned model provided direct, succinct answers where the base model offered lengthy preambles.

The qualitative improvements in Gemma-2-9B and Qwen2-7B were less pronounced, aligning with their lower overall improvement scores. However, Qwen2-7B did show improvements in conciseness, with fine-tuned responses being up to 58% shorter while maintaining information quality.

These results demonstrate that fine-tuning effectiveness varies considerably across model architectures, with smaller models like Llama-3.2-3B showing unexpectedly large improvements in healthcare-specific response quality. This suggests that model size is not the sole determinant of fine-tuning potential, and carefully tuned smaller models may offer competitive performance in specialized domains like healthcare [8].

VI. Conclusions

This study demonstrates the significant potential of fine-tuning small LLMs for enhancing the performance of AI healthcare chatbots. Our comprehensive analysis across multiple model architectures reveals several key findings with important implications for deploying these models in healthcare settings.

First, our results show that model size is not necessarily the primary determinant of fine-tuning effectiveness [8]. The smallest model in our study, Llama-3.2-3B, demonstrated the most substantial overall improvement (108.2 points), particularly in direct answer capabilities, where it showed a remarkable 30 percentage point increase. This suggests that carefully fine-tuned smaller models can achieve competitive performance in specialized domains while potentially requiring fewer computational resources.

Second, we observed distinct patterns of improvement across different model architectures. Mistral-7B exhibited the most significant efficiency gains, reducing generation time by 22.24% while simultaneously improving direct answer capabilities by 20 percentage points. In contrast, Gemma-2-9B and Qwen2-7B showed more modest improvements, suggesting that model architecture plays a crucial role in determining fine-tuning outcomes.

Third, our analysis reveals trade-offs between different aspects of model performance. While Llama-3.2-3B improved substantially in response quality and structure, it experienced a slight increase in generation time. This highlights the importance of considering multiple performance dimensions when selecting and fine-tuning models for specific healthcare applications.

The structural changes in responses after fine-tuning—particularly the increased use of numbered lists in Llama-3.2-3B and the reduction in overly formatted text across most models—suggest that fine-tuning can effectively align output formats with domain-specific conventions. This is particularly valuable in healthcare contexts, where clear, structured information presentation is critical.

Our approach of using synthetic data generation supplemented with domain-specific questions proved effective for creating a robust training dataset. This methodology may be particularly valuable in specialized domains like healthcare, where obtaining large volumes of expert-annotated data can be challenging and costly.

Future work should explore the integration of retrieval-augmented generation (RAG) techniques to further enhance factual accuracy and currency of information, which is particularly crucial in rapidly evolving healthcare domains [7]. Additionally, investigating the generalizability of these fine-tuned models across different healthcare subdomains and evaluating their performance with diverse patient populations would provide valuable insights for real-world deployment.

In conclusion, our findings demonstrate that fine-tuning, particularly when applied to smaller LLMs like Llama-3.2-3B, offers a promising approach for developing more accurate, efficient, and responsive AI healthcare chatbots. These improved models have the potential to enhance patient education, support healthcare providers, and contribute to more accessible health information systems [2], [9], [13].

References

[1] A. Vaswani *et al.*, "Attention is all you need," *arXiv (Cornell University)*, Jan. 2017, doi: 10.48550/arxiv.1706.03762.

[2] H. M. Alghamdi and A. Mostafa, "Towards Reliable Healthcare LLM Agents: A Case Study for Pilgrims during Hajj," *Information*, vol. 15, no. 7, p. 371, Jun. 2024, doi: 10.3390/info15070371.

[3] A. Azam, Z. Naz, and M. U. G. Khan, "PharmaLLM: A medicine prescriber chatbot exploiting Open-Source large language models," *Human-Centric Intelligent Systems*, Nov. 2024, doi: 10.1007/s44230-024-00085-z.

[4] "Comparative performance analysis of large language models for deployment in women's healthcare mobile applications," *IEEE Conference Publication | IEEE Xplore*, Nov. 21, 2024. https://ieeexplore.ieee.org/document/10867016

[5] B. Lajčinová, P. Valábek, and M. Spišiak, "Intent Classification for Bank Chatbots through LLM Fine-Tuning," *arXiv (Cornell University)*, Oct. 2024, doi: 10.48550/arxiv.2410.04925.

Authorized licensed use limited to: The University of Toronto. Downloaded on August 05,2025 at 10:52:58 UTC from IEEE Xplore. Restrictions apply.

[6] P. Vadlapati, "AutoPureData: Automated filtering of undesirable web data to update LLM knowledge," *Journal of Mathematical & Computer Applications*, pp. 1–4, Aug. 2024, doi: 10.47363/jmca/2024(3)e121.

[7] H. K. Chaubey, G. Tripathi, R. Ranjan, and S. K. Gopalaiyengar, *Comparative analysis of RAG, Fine-Tuning, and Prompt Engineering in Chatbot development*. 2024, pp. 169–172. doi: 10.1109/icftss61109.2024.10691338.

[8] D. M. Anisuzzaman, J. G. Malins, P. A. Friedman, and Z. I. Attia, "Fine-Tuning LLMs for specialized use cases," *Mayo Clinic Proceedings Digital Health*, vol. 3, no. 1, p. 100184, Nov. 2024, doi: 10.1016/j.mcpdig.2024.11.005.

[9] A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez, T. F. Tan, and D. S. W. Ting, "Large language models in medicine," *Nature Medicine*, vol. 29, no. 8, pp. 1930–1940, Jul. 2023, doi: 10.1038/s41591-023-02448-8.

[10] Y. Shen *et al.*, "ChatGPT and other large language models are double-edged swords," *Radiology*, vol. 307, no. 2, Jan. 2023, doi: 10.1148/radiol.230163.

[11] H. Touvron *et al.*, "LLAMA: Open and Efficient Foundation Language Models," *arXiv (Cornell University)*, Jan. 2023, doi: 10.48550/arxiv.2302.13971.

[12] A. L. Beam and I. S. Kohane, "Big data and machine learning in health care," *JAMA*, vol. 319, no. 13, p. 1317, Mar. 2018, doi: 10.1001/jama.2017.18391.

[13] M. Sharma, C. Savage, M. Nair, I. Larsson, P. Svedberg, and J. M. Nygren, "Artificial intelligence Applications in Health care Practice: Scoping review," *Journal of Medical Internet Research*, vol. 24, no. 10, p. e40238, Aug. 2022, doi: 10.2196/40238.

[14] "mlx-community/gemma-2-9b-it-4bit · Hugging Face," Oct. 22, 2024. https://huggingface.co/mlx-community/gemma-2-9b-it-4bit

[15] "mistralai/Mistral-7B-Instruct-v0.3 · Hugging Face." https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3

[16] "mlx-community/Qwen2-7B-Instruct-8bit · Hugging Face." https://huggingface.co/mlx-community/Qwen2-7B-Instruct-8bit

[17] "phi4," *Ollama*. https://ollama.com/library/phi4

[18] E. J. Hu *et al.*, "LORA: Low-Rank adaptation of Large Language Models," *arXiv.org*, Jun. 17, 2021. https://arxiv.org/abs/2106.09685

Authorized licensed use limited to: The University of Toronto. Downloaded on August 05,2025 at 10:52:58 UTC from IEEE Xplore. Restrictions apply.